

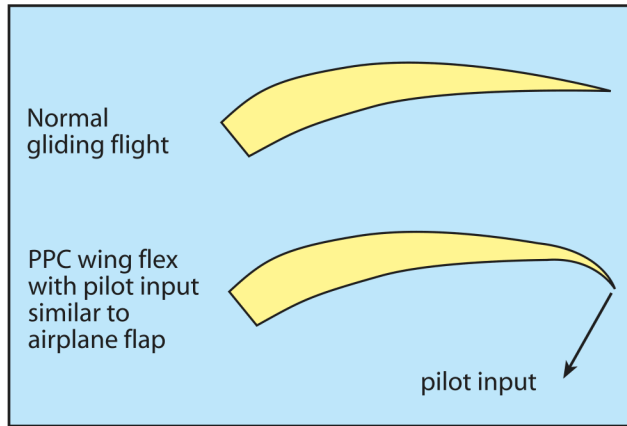


Figure 2-9. PPC wing flexibility in flight.

3. The sum of all moments equals zero.

THRUST – the forward force produced by a power-plant/propeller as it forces a mass of air to the rear (usually said to act parallel to the longitudinal axis).

VS.

DRAW – the aerodynamic force acting on the airfoil lines and cart in the same plane and in the same direction as the relative wind.

LIFT – the aerodynamic force caused by air flowing over the wing that is perpendicular to the relative wind.

VS.

WEIGHT – the force of gravity acting upon a body.

Lift

Lift opposes the downward force of weight and is produced by the dynamic effects of the surrounding air-stream acting on the wing. Lift acts perpendicular to the flight path through the wing's center of lift. There is a mathematical relationship between lift, angle of attack, airspeed, altitude, and the size of the wing. In the lift equation, these factors correspond to the terms coefficient of lift, velocity, air density, and wing surface area. The relationship is expressed in Figure 2-11.

This shows that for lift to increase, one or more of the factors on the other side of the equation must increase. Lift is proportional to the square of the velocity, or airspeed, therefore, doubling airspeed quadruples the amount of lift if everything else remains the same. Small changes in airspeed create larger changes in lift. Likewise, if other factors remain the same while the coefficient of lift increases, lift also will increase. The coefficient of lift goes up as the angle of attack is increased. As air density increases, lift increases. However, you will usually be more concerned with

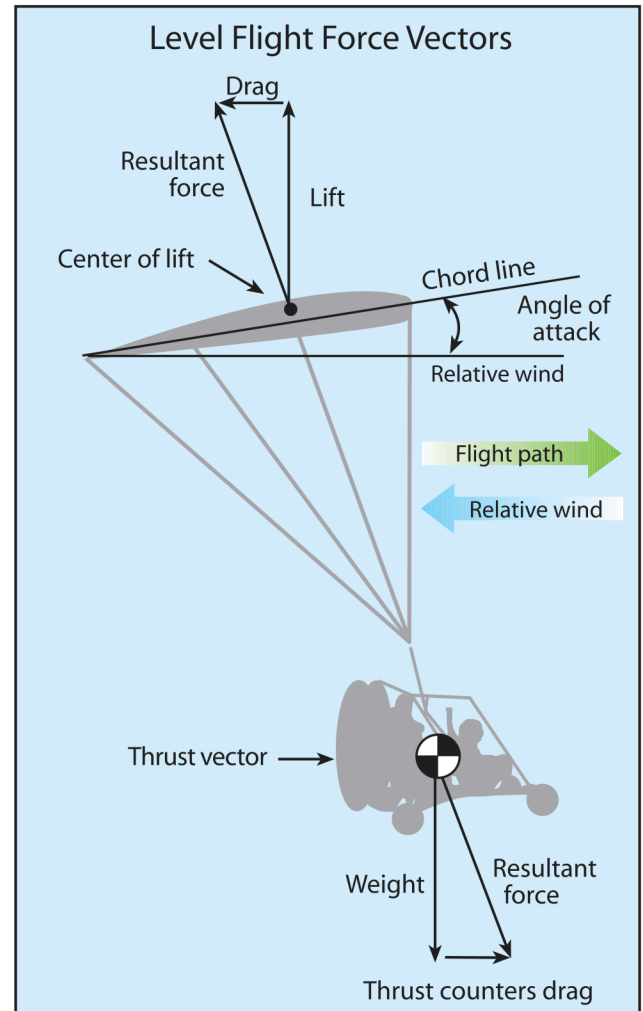


Figure 2-10. Level flight forces.

$$L = C_L V^2 \frac{\rho}{2} S$$

L = Lift

C_L = Coefficient of lift

(This dimensionless number is the ratio of lift pressure to dynamic pressure and area. It is specific to a particular airfoil shape, and below the stall, it is proportional to angle of attack.)

V = Velocity (Feet per second)

ρ = Air density (Slugs per cubic foot)

S = Wing surface area (Square feet)

Figure 2-11. The lift equation.